

CS 556: Task-Sheet for Lab 4

Wall Following (P-Controller)

Name: _____ team skynet ultron _____

Robot ID: _____ 19 _____

Question:	1a Prelab	1b Prelab	1c Prelab	3.1	3.2a	3.2b	3.2c	3.2 d	3.2e	Total
Points:	5	5	10	30	26	6	6	6	6	100
Score:										

Pre-lab Submission (on Canvas, Due Wednesday, 23:59 PM before lab #4): • 1)

Task-Sheet Pre-lab task (one PDF file: [TeamX_LabY_Prelab_Tasksheet.pdf](#)) • 2)

Draft code (one zip folder: [TeamX_LabY_DraftCodes.zip](#))

Post-lab submission (On Canvas, Due Wednesday, 23:59 PM after lab #4) Submit in 3 separate files (do NOT zip these four together):

- 1) Task-Sheet tasks (one PDF file: [TeamX_LabY_Postlab_Tasksheet.pdf](#)).
- 2) Final code (one ZIP folder: [TeamX_LabY_FinalCodes.zip](#)).
- 3) Videos (one zip folder: [TeamX_LabY_Videos.zip](#)).

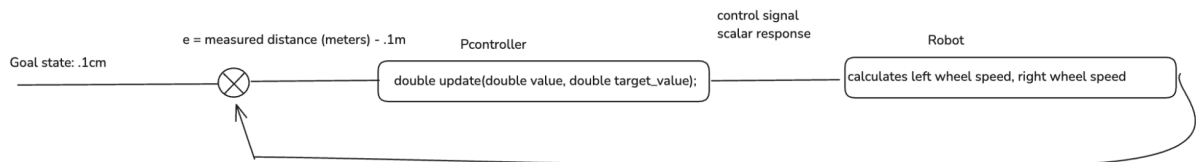
Lab Grading Criteria:

- Every member of the team must submit the same set of final materials for the post-lab to receive a grade.
- For the pre-lab submissions, no extensions will be offered. Failure to prepare for the lab will result in a 20% deduction from your total grade for that week's lab.
- For the post-lab submissions, late work or post-lab submissions will incur a 20% penalty per day, up to a maximum deduction of 100%.
- Lab attendance is mandatory – a missed lab will mean 0 points for you.
- Before you leave, make sure you return all robots and tools used to the front desk in the same format that you received them. If not done so, 10 points will be deducted.
- Your code must follow the industry guidelines for coding; if not done so, 10 points will

be deducted. Comment out code for easy review, add new code below, and label each task with clear, descriptive comments. Ensure a modular, well-structured design with meaningful names, minimal globals, and avoid hardcoding values.

1. Prelab

- a. (5 points) Draw the block diagram for your proportional PController. Make sure to label and give units for the goal state, error, and controller output.



- b. (5 points) (Controller Architecture)

i. Current state and how to estimate it:

_____ use the measurement from the sonar

ii. Goal state:

_____ fiven to be .1m clamped with upper and lower bounds _____

iii. Function for right wheel:

_____rightSpeed = base_speed -

Pout;_____

iv. Function for left wheel:

_____rightSpeed = base_speed +

Pout;_____

c. (10 points) PController Draft Code

3.1 (30 points) Final PController class submission

3.2 PController Wall Following

- a. (26 points) Wall Following Working [Video for each test (b-d), **3 videos in total**] b. (6 points) List the first set of gain K_p and clamping ranges, and describe the robot's behavior.

- c. (6 points) List the second set of gain K_p and clamping ranges, and describe the robot's behavior.

- d. (6 points) List the third set of gain K_p and clamping ranges, and describe the robot's behavior.

e. (6 points) Best gain K_p _____